

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of BSc Biotechnology/ Microbiology/ Biochemistry Examination March 2018

Course code & Name: BS501 Basic Molecular Biology-II

Date: 26.03.2018 Day: Monday Time: 01:30 PM To 01:50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

10

1. Molecular biology is the study of
 - (i) DNA, RNA and Proteins
 - (ii) Lipids and Carbohydrates
 - (iii) Carbohydrates only
 - (iv) None of the above
2. Chromatin is an association of
 - (i) DNA and proteins
 - (ii) DNA and carbohydrates
 - (iii) DNA and lipids
 - (iv) All the above
3. Transcription is formation of
 - (i) RNA from DNA
 - (ii) DNA from DNA
 - (iii) DNA from RNA
 - (iv) RNA from RNA

4. Which if the following is not a stop codon
(i) TAA (ii) TGA (iii) TAG (iv) TTT
5. Ribosome is made up of
(i) rRNA (ii) tRNA
(iii) DNA (iv) All the above
6. Translation is formation of
(i) DNA from Protein (ii) Protein from RNA
(iii) Protein from Protein (iv) RNA from Protein
7. Full-form of DNA is
(i) Doxyribonucleic acid (ii) Deoxyribonucleic acid
(iii) Dideoxyribonucleic acid (iv) Dioxyribonucleic acid
8. Initiation, elongation and termination steps are associated with
(i) DNA replication (ii) Translation
(iii) Transcription (iv) All the above
9. Shine Dalgarno sequence is found on
(i) mRNA (ii) rRNA (iii) miRNA (iv) snoRNA
10. Anti-codon sequences are found on
(i) tRNA (ii) miRNA
(iii) mRNA (iv) DNA

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V Semester of BSc Biotechnology/ Microbiology/ Biochemistry Examination March 2018

Course code & Name: BS501 Basic Molecular Biology-II

Date: 26.03.2018 Day: Monday Time: 01:51 AM To 03:30 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed **10**

1. Mention any five differences between prokaryotic and eukaryotic transcription 5
OR
Describe prokaryotic transcription
2. Describe prokaryotic translation 5
OR
Describe eukaryotic translation

SECTION – II

Q-III Answer the following questions as directed **15**

1. What do you understand by Chromatin? Discuss its various forms 5
OR
Describe eukaryotic transcription
2. Describe a typical structure ribosome 5
OR
Discuss unique and repetitive DNA
3. Write a short note on **ANY TWO** 5
(i) lac operon
(ii) Molecular switch of phage lambda
(iii) Wobble Hypothesis
(iv) Linking number